

Tiny Society AI: Calculations and Measurements

A short reference for the numbers shown during a run

Tiny Society AI | 19 September 2026

This guide keeps the useful parts: what each number means, how the engine calculates it, and which constants matter for tuning. One simulated day contains morning, afternoon, evening, and night. Relationship and stance values live in $[-1, 1]$, influence in $[-100, 100]$, and memory importance in $[1, 10]$.

1 Daily social metrics

Metric	Meaning	Calculation
friendship_count, rivalry_count, conflict_count, romance_count, alliance_count	How many pairs have each relationship type	Count each unordered pair once if either direction has that type.
average_relationship_strength	Average bond intensity	Mean $ strength $ over directed relationship edges.
average_trust_score	Warmth among positive trust edges	Mean positive trust strengths; returns 0 when none exist.
most_connected	Characters with the most social links	Top 3 by outgoing relationship count.
relationship_volatility	Story-relevant relationship changes today	Milestones emitted by the consequence layer.
social_fragmentation	Share of isolated characters	Agents with no outgoing relationships divided by population.
group_centrality	Groups that bridge outward	Top 5 groups by cross-group relationship edges.
influence_gainers, influence_losers	Who rose or fell since setup	Top/bottom 3 influence changes against Day 0.

2 Beliefs and forecast confidence

Each agent may hold a stance on a topic: $s_{i,q} \in [-1, 1]$. For each topic q , the engine records

$$\mu_q = \text{mean}(s_{i,q}), \quad \sigma_q = \text{population standard deviation}(s_{i,q}).$$

The forecast confidence is

$$\Gamma = \text{clamp}(1 - \text{mean}_q(\sigma_q), 0, 1).$$

High confidence means the agents agree with each other. It is consensus, not proof the forecast is correct.

The final forecast copies the final day's topic means, uncertainties, and confidence. Pivotal days are the three days with the largest positive total movement:

$$\text{drift}(t) = \sum_q |\mu_q(t) - \mu_q(t-1)|.$$

3 Memory retrieval

New memories get a cheap importance score:

$$\text{importance} = \text{clamp}(1 + \min(\text{chars}/60, 4) + \min(1.5 \cdot \text{keywordHits}, 5), 1, 10).$$

Reflections use 9.0; direct player guidance uses 10.0.

When an agent recalls memories, each candidate combines relevance, recency, and importance:

$$\begin{aligned} \text{relevance} &= \text{stopword-filtered token overlap with the query,} \\ \text{recency} &= 0.85^{\text{daysSinceLastAccess}}, \\ \text{score} &= \text{norm}(\text{relevance}) + \text{norm}(\text{recency}) + \text{norm}(\text{importance}/10). \end{aligned}$$

A diversity penalty of 0.6 times token overlap with already selected memories keeps recall from returning several near-duplicates. Retrieved memories become recent again.

4 Feed visibility and ranking

Action kind	Who sees it
post	Everyone.
direct	Actor and named targets.
comment, interact, amplify	Actor, targets, group-mates, and everyone if actor influence is at least 20.0.

Each agent keeps the latest 15 feed entries. Ranked feeds use

$$score = 3.0 \cdot interestMatch + 1.0 \cdot normalizedAuthorInfluence + 1.5 \cdot recency, \quad recency = 0.85^{daysSinceEntry}.$$

Interest comes from overlap with the viewer’s traits, revealed traits, stance topics, relationship names, and own name.

5 Choosing who acts

Each day the engine scores agents, then softmax-samples without replacement. This is not a strict top-N list, so quieter agents can still get turns.

$$\begin{aligned} selectionScore = & 3.0 \cdot \mathbf{1}[\text{event mentions agent traits, goals, groups, or name}] \\ & + 2.0 \cdot influence / maxInfluence \\ & + 1.5 \cdot \mathbf{1}[\text{charged relationship}] \\ & + 0.5 \cdot \min(groupCount, 3). \end{aligned}$$

Charged relationships are rivalry, romance, or conflict with $|strength| \geq 0.3$. The softmax temperature is 1.0. The default daily reasoning limit is 8 agents.

6 Relationship updates

Agents choose social intents such as support, flirt, confront, or ally. The code maps those intents to fixed affinity bids; the model does not get to invent relationship strengths directly.

$$appliedAffinityChange = \text{clamp}(intentBid, -0.4, 0.4) \cdot 0.5.$$

Type	First qualifying rule
romance	Both sides have affinity at least 0.50, romantic signal at least 0.40, and at least 4 interactions.
alliance	Both sides have affinity at least 0.35, alliance signal at least 0.40, and at least 3 interactions.
conflict	This side’s affinity is at most -0.50 .
rivalry	This side’s affinity is at most -0.25 .
friendship	This side’s affinity is at least 0.30.
trust	This side’s affinity is at least 0.10.
unchanged	Near-neutral affinity keeps the previous label.

Romance and alliance require mutual evidence. Rivalry and conflict can be one-sided.

7 Influence, stance, and background drift

Actors gain base influence by action kind: post +0.6, amplify +0.3, comment/interact +0.2, direct +0.1. If the action antagonizes at least one target, the actor gets another +0.3 and each antagonized target gets -0.4 . Amplifying someone also gives that target +2.0. Influence is clamped to $[-100, 100]$.

Action stance shifts are applied only to the actor:

$$stance'_q = \text{clamp}(stance_q + \text{clamp}(\delta a_q, -0.3, 0.3), -1, 1).$$

Agents who do not reason with the language model still drift:

Rule	Update
Warm social bonds	Social, charismatic, loyal, or patient agents add +0.02 to positive bonds.
Ordinary bond fade	Positive bonds lose 0.01; negative bonds move +0.01 toward neutral.
Isolation penalty	Introverted, anxious, or lonely agents with no relationships lose 0.1 influence, floored at -10 .
Shared-group trust	Each shared-group pair has an 8% chance to gain +0.01 both ways.
Group stance drift	Agents move 2% toward their group-mates' mean stance; contrarian traits move away.
Influence regression	Nonzero influence is multiplied by 0.99.

8 Main tuning constants

Constant	Value	Constant	Value
Daily reasoning agents	8	Reflection interval	4 days
Plan refresh horizon	3 days	Exchange-turn cap	2 extra turns
Exchange-chain cap	3/day	Vignette cap	2/day
Dynamic event cadence	every 5 days after Day 3	Observation window	15
Public influence threshold	20.0	Memory/feed recency decay	0.85
Recall diversity penalty	0.6	Affinity damping	0.5
Romance/alliance signal step	0.25	Final pivotal days	3
LLM concurrency	6 calls		

9 Caveat

These are model measurements, not measurements of real societies. They are useful for debugging and comparing simulation runs, but claims about real people would need separate validation against real-world data.